Changing velocity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate acceleration and distinguish speeding up, slowing down, and turning.

SUCCESS CRITERIA - I can...

1. Calculate acceleration from velocity change and time.
2. Compare acceleration direction with velocity direction.
3. Explain why turning at steady speed is acceleration.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “acceleration” and “initial” to compare your model or result. Sentence starters are optional.

acceleration: rate of change of velocity

initial: at the beginning of the interval; subscript i

final: at the end of the interval; subscript f

01 Bring back an earlier idea

Velocity is +4 m/s for 6 s, then -3 m/s for 4 s.
Find displacement and distance.
Explain the reference information needed for an angle of 30°.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a_{avg} = (v_f - v_i)/\Delta t$.
i = initial; f = final. Compare directions of velocity and acceleration to decide whether speed increases.

Worked reasoning: A cart moving east slows from 8 to 2 m/s in 3 s. Taking east positive, $a=(2-8)/3=-2$ m/s². It keeps moving east while acceleration points west.

Your turn: Draw initial and final velocity arrows and one acceleration arrow.
Use arrow lengths to show the speed change.

Sentence starter: The ... in my model represents ... . Its label shows ... .

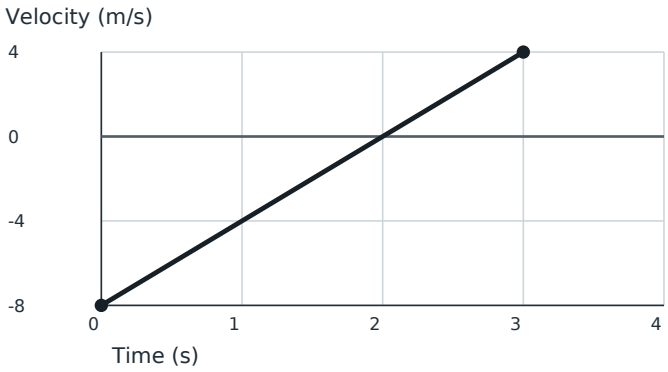
Marzano check - circle: ① ② ③

03 Signed acceleration

Use the graph to calculate acceleration and explain the speed trend.
State when the cart is momentarily at rest.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: ① ② ③

Changing velocity

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Braking intervals

A car travels at +24 m/s and slows at a constant rate to 0 m/s in 7.0 s. The positive direction is the car's original direction of travel.
(a) Calculate its acceleration.
(b) Calculate its displacement during braking.
(c) Explain the sign of each answer.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

05 Small acceleration, long time

A hypothetical constant sideways acceleration is 2×10^{-6} m/s² for 10^{6} s.
Find Δv and explain why small acceleration alone cannot justify ignoring it.

Word help: justify = support your answer with a reason or evidence

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A=(5, -3) m and B=(-3, 7) m.
Find R, magnitude and direction.
Choose C so A+B+C=0 and sketch the route.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

Velocity changes from -3 to -9 m/s in 4 s, then stays -9 m/s for 4 s.
Find acceleration and speed trend in each interval.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A cart begins at -6 m/s and has constant acceleration +2 m/s² for 5 s.
Sketch v versus t; find turning time, displacement and distance.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3